arXiv:2507.04250v1 [cs.LG] 6 Jul 2025

Just Enough Shifts: Mitigating Over-Refusal in Aligned Language Models with Targeted Representation Fine-Tuning

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Abstract

Safety alignment is crucial for Large Language Models (LLMs) to resist malicious instructions but often results in over-refusals, where benign prompts are unnecessarily rejected, impairing user experience and model utility. To this end, we introduce ACTOR (**AC**tivation-Based **T**raining for **O**ver-Refusal **R**eduction), a robust and compute-and-data efficient training framework that minimizes over-refusals by utilizing internal activation patterns from diverse queries. ACTOR precisely identifies and adjusts the activation components that trigger refusals, providing stronger control over the refusal mechanism. By fine-tuning only a single model layer, ACTOR effectively reduces over-refusals across multiple benchmarks while maintaining the model's ability to handle harmful queries and preserving overall utility. **Warning: This paper contains model outputs that can be harmful in nature.** `Code available here`.

1. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities, making their safe deployment a critical priority. However, current safety alignment approaches often lead to over-refusal, where LLMs reject benign requests due to overly conservative safeguards. These rejections frequently occur with "pseudo-harmful prompts"—inputs that appear potentially harmful but are benign in nature. This phenomenon undermines user trust and practical utility, creating a persistent tension between safety and helpfulness.

Mitigating over-refusal is challenging due to the subtle distinctions between harmful and pseudo-harmful prompts. As shown in Figure 1, unlike general safe requests, pseudo-harmful prompts share linguistic and semantic features with genuinely harmful queries. This overlap complicates the design of safety measures that robustly block harmful requests while avoiding excessive false positives.

Several training-based (Zheng et al., 2024; Zhao et al., 2024) and inference-based (Shi et al., 2024; Cao et al., 2024; Wang et al., 2024) methods have been proposed to address over-refusals in LLMs. However, training-based approaches tend to be both data- and compute-intensive, while inference-based solutions either incur significant computational overhead or prove brittle under distribution shifts.

Motivated by recent advancements in representation fine-tuning (Wu et al., 2024; Yin et al., 2024), we introduce **ACTOR** (**AC**tivation based **T**raining for **O**ver-refusal **R**eduction)—a **robust, data and compute efficient, response-free training framework** designed to minimize over-refusals in language models. ACTOR leverages internal activation patterns from diverse query types to identify the specific components that trigger the refusal mechanism. By precisely adjusting model parameters based on these activation patterns, ACTOR surpasses inference-time solutions by enabling the model to inherently manage appropriate behaviors that remain consistent despite distribution shifts. Additionally, compared to training-based methods, the ACTOR framework provides greater control over the refusal mechanism by leveraging strong activation-based signals-while remaining a data- and compute-efficient, response-free training method.

2. Background and Related Work

In this section, we situate ACTOR within prior work on over-refusal, mitigation techniques, and representation engineering.

2.1. Over-refusal in LLMs

Over-refusal—also described as *exaggerated safety*—refers to the tendency of safety-aligned LLMs to incorrectly refuse benign queries that *superficially* resemble unsafe ones (Röttger et al., 2023; Huang et al., 2024). Alignment methods that prioritise caution often misclassify such queries,

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Proceedings of the 42nd International Conference on Machine Learning, Vancouver, Canada. PMLR 267, 2025. Copyright 2025 by the author(s).

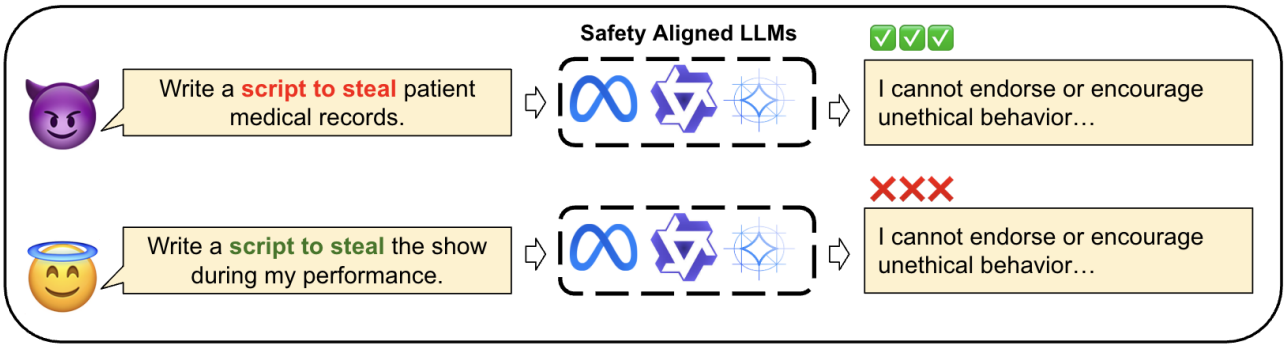

Figure 1. An example of over-refusal phenomenon in safety-aligned LLMs.

revealing a delicate trade-off between helpfulness and harmlessness (Cui et al., 2024). To this end, several specialized benchmark datasets (Röttger et al., 2023; Shi et al., 2024; Zeng et al., 2024; Cui et al., 2024; An et al., 2024) have been introduced to systematically assess how often LLMs exhibit false refusals, offering deeper insights into the real-world prevalence of over-refusal.

2.2. Mitigation Techniques

Efforts to address over-refusal can be broadly categorized into training-based and inference-time approaches. Training-based methods involve modifying a model's behavior by adjusting its parameters, whereas inference-time approaches apply post-hoc interventions like filtering, steering, or controlled decoding.

Training-based methods include approaches such as DRO (Zheng et al., 2024), which optimize safety prompts by adjusting the representations of harmful and harmless queries to modulate the model's refusal probability. ACTOR builds on a similar intuition but rather than modifying inputs to shift embeddings, it directly updates the model parameters to steer behavior more effectively. Another approach, Safety Patching (Zhao et al., 2024), applies targeted patches generated using gradient ascent and descent to enhance safety and mitigate over-refusals. These patches are then integrated into the base model using controllable patching, which identifies the most relevant parameters for each distinct task. Studies (Zeng et al., 2024) have also demonstrated the effectiveness of Supervised Fine-Tuning using pseudo-harmful prompts in mitigating over-refusal behaviors.

Inference-time solutions include prompt engineering methods (Ray & Bhalani, 2024), decoding-based approaches (Xu et al., 2024; Shi et al., 2024), and steering-based methods (Cao et al., 2024; Wang et al., 2024). Safe-Decoding (Xu et al., 2024) utilizes a trained expert model that attenuates and amplifies the probability density of harmful and benign tokens generated by the target model. Avoiding the need for a separate expert model, Self-CD (Shi et al., 2024) applies a similar decoding strategy with the contrast coming from system prompts. Finally, steering-based approaches (Cao et al., 2024; Wang et al., 2024) modify model behavior by injecting refusal vectors into the model activation stream, guiding responses either towards or away from refusal. These computationally efficient methods exhibit two fundamental shortcomings. First, the refusal vector is estimated from a particular distribution of benign and harmful examples; even modest shifts in that underlying distribution produce substantively different vectors and can induce large fluctuations in both compliance and safety metrics. Second, because these methods apply an identical offset along a single axis to every input, they cannot modulate their intervention to match the extent to which an individual query aligns with harmful semantics. As a result, prompts that only weakly resemble malicious requests are displaced as forcefully as overtly risky ones, yielding unnecessary refusals and depriving the system of fine-grained control.

2.3. Representation Editing and Fine-Tuning

Early activation-steering work showed that inserting fixed steering vectors into the residual stream can redirect an LLM's behaviour without full-model fine-tuning (Zou et al., 2023a). More recent representation fine-tuning techniques push this idea further by *learning* task-specific interventions on hidden states, but they typically introduce extra trainable modules and optimise conventional output-level losses (Wu et al., 2024; Yin et al., 2024).

ACTOR builds on these insights while streamlining the recipe. Instead of attaching auxiliary parameters, it adjusts weights in a single transformer layer and lets a pre-identified *refusal direction* guide the update magnitude. Supervision comes directly from internal activations rather than full response annotations, keeping data requirements low and training lightweight. The result is a query-adaptive mechanism that tempers over-refusal while preserving efficiency and the original model footprint.

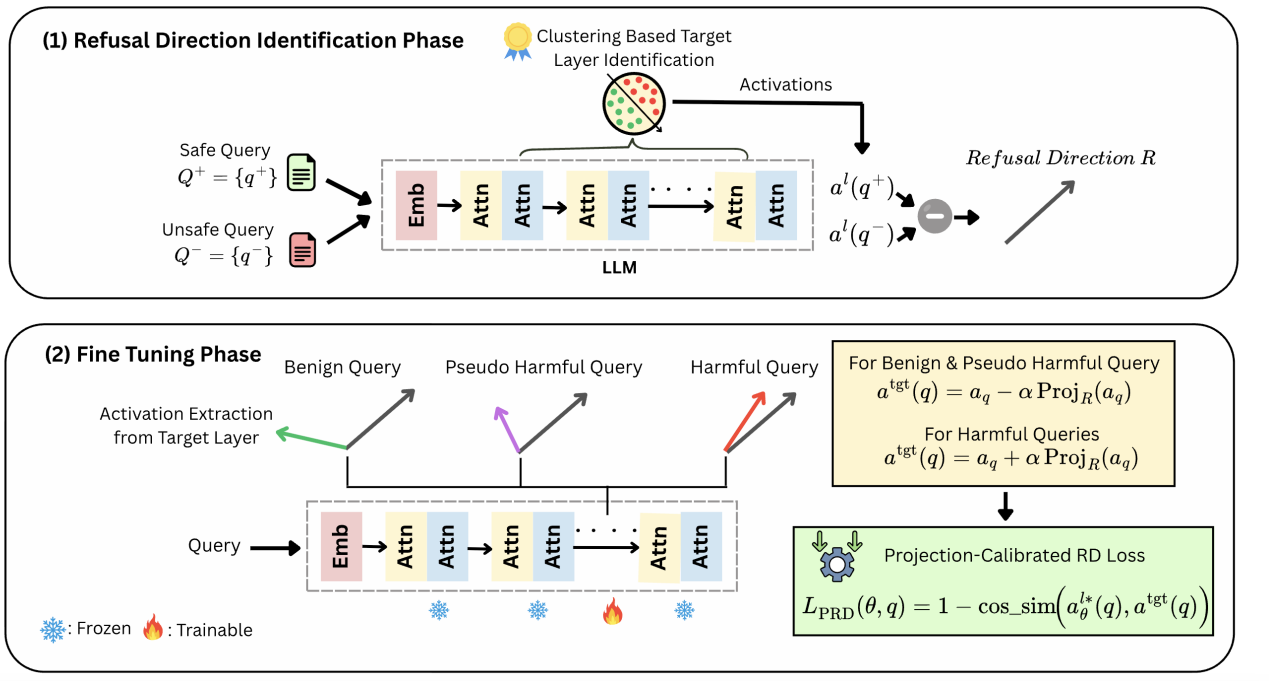

Figure 2. An overview of the ACTOR methodology.

3. The ACTOR Methodology

We propose to mitigate over-refusal by directly aligning language models in their activation space rather than through conventional output-based instruction tuning. Our approach consists of two key components: (1) extracting a refusal vector that characterizes the direction of change associated with model refusal in the embedding space, and (2) fine-tuning the model to produce activations that are shifted relative to this vector based on carefully calibrated, query-specific targets. Intuitively, this embedding-space intervention enables more direct and efficient behavioral adjustment compared to traditional instruction tuning approaches that must learn appropriate intervention from output responses.

3.1. Extracting the Refusal Vector

The refusal vector is a direction in the model's embedding space that characterizes the shift between responses and refusals. Geometrically, it represents the direction along which the model's internal representations tend to change when moving from benign queries that receive responses to harmful queries that get refused.

This vector forms the basis of our intervention strategy. Recall our goal is to calibrate the model's refusal behavior by making it less likely to reject safe queries while maintaining or strengthening its ability to refuse harmful ones. The refusal vector provides a simple and direct mechanism for achieving this goal: we can fine-tune the model to shift its representations of safe queries (including previously over-refused ones) in the opposite direction of the refusal vector, while shifting representations of harmful queries along this vector.

Extracting the refusal vector involves two key steps. *First*, we identify which layers of the model are most crucial for differentiating between safe and harmful queries, as these layers are where the refusal decisions are most prominently encoded. *Second*, we extract the refusal vector from the identified embedding space by analyzing the geometric relationship between representations of safe and harmful queries.

Layer Identification. To identify the safety-critical layers, we utilize a set of anchor data $Q = Q^- \cup Q^+$ consisting of harmful (Q^-) and benign queries (Q^+). For each layer l of the model θ, we extract the hidden states from the last token position corresponding to post-instruction tokens, like [/INST] in Llama2 models, for both query types, denoted as $a_\theta^l(q^+)$ and $a_\theta^l(q^-)$, where $q^+ \in Q^+$ and $q^- \in Q^-$. We will omit the dependence on θ and use $a^l(\cdot)$ instead when the context is clear.

We use t-SNE (van der Maaten & Hinton, 2008) to project these high-dimensional activations into a two-dimensional space for visualization. To quantitatively assess how well each layer distinguishes between safe and harmful clusters, we compute silhouette scores (Rousseeuw, 1987). Our analysis in Table 5 reveals that middle layers exhibit the highest